

Java Programming Assignment

Data Types, Variables & Operators

Q1

Which of the following is NOT a primitive data type in Java?

- A. int
- B. boolean
- C. String
- D. char

Q2

What will be printed?

```
int x = 7;  
int y = 2;  
System.out.println(x / y);
```

- A. 3
- B. 3.5
- C. 4
- D. Compilation Error

Q3

Which declaration is valid?

- A. float price = 19.99;
- B. float price = 19.99f;
- C. float price = "19.99";
- D. float price = '19.99';

must use f as decimal nums considered as double

Q4

What is the value of x?

```
int x = 8;  
x %= 3;
```

- A. 0
- B. 1
- C. 2
- D. 3

Q5

Which expression evaluates to false?

- A. `5 == 5`
- B. `10 >= 5`
- C. `8 != 8`
- D. `7 < 9`

Q6

Which operator has the highest precedence?

- A. `+`
- B. `*`
- C. `==`
- D. `&&`

Q7

Which variable name follows Java naming conventions?

- A. `student-name`
- B. `2number`
- C. `studentName`
- D. `class`

Q8

What is the output? 20

```
System.out.println(10 + 5 * 2);
```

Q9

Predict the output.

```
int x = 5;
System.out.println(x++); 5
System.out.println(x);    6
```

Q10

Predict the output.

```
int x = 5;
System.out.println(++x); 6
System.out.println(x);    6
```

Q11

Determine the output.

```
int a = 10;
int b = 3;
System.out.println(a / b); 3
System.out.println(a % b); 1
```

Q12

Predict the output.

```
boolean a = true;
boolean b = false;
System.out.println(a || b && false);
```

 true

Q13

What will be printed?

```
int x = 2, y = 3, z = 4;
System.out.println(x + y * z - 1);
```

 13

Q14

```
int x = 10;
System.out.println(x++ + ++x);
```

 10 + 12

Q15

```
int x = 4;
System.out.println(x++ * 2 + --x);
```

 4 4 12

Q16

What is the problem?

```
int age = "20";
```

- A. No error
- B. Wrong data type assigned
- C. Missing semicolon
- D. Invalid variable name

Q17

Identify the error.

```
boolean passed = "true";
```

- A. No error
- B. "true" should be true
- C. Use single quotes
- D. boolean cannot store true/false

Q18

Will this code compile?

```
int x;
x = x + 5;
```

```
System.out.println(x);
```

- A. Yes, it prints 5
- B. Yes, it prints 0
- C. Compilation Error declared not initialized
- D. Runtime Error

Bonus

Without running the code, determine the output and explain your answer.

```
int a = 5;  
int b = 10;  
System.out.println(a++ + b-- + ++a + --b);
```

a++ =5 -> now a=6
b-- =10 -> now b=9
++a=7
--b =8
so 5+10+7+8 =30